

p-adic multiple L-functions and twisted multiple Bernoulli numbers

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- ① Overview
- ② Generalization of Bernoulli numbers
- ③ p-adic multiple L-functions
- ④ The case of depth one
- ⑤ References

Overview

Furusho and Jarossay's work

Theorem

For any r -tuple $(n_i)_r$ of positive integers, the family

$$\left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{l=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \sum_{\substack{\xi \in \mu_c \\ \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2, 3)}}{\epsilon_j} \right)^{-\kappa'_j} \\ \xi = \epsilon}} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \mathcal{B}_{0, \xi}^{(l_J, \epsilon_J, \kappa_J)} \sum_{\delta \in \Delta(T_{r,J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}.$$

$$p\text{-adic } L\text{-functions} = \sum (\text{a variant of CMBNs}) \times (\text{CMHVs})$$

Furusho, Komori, Matsumoto, and Tsumura's work

Theorem

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity and $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$). Assume that

$$\xi_j \neq 1 \text{ for all } j \ (1 \leq j \leq r).$$

Then, with the above notation, the generalized Euler–Zagier–Lerch type complex multiple zeta functions $\zeta_r((s_j); (\xi_j); (\gamma_j))$ can be analytically continued to $\mathbb{C}^r$ as an entire function in (s_j) . For $n_1, \dots, n_r \in \mathbb{N}_0$,

$$\zeta_r((-n_j); (\xi_j); (\gamma_j)) = (-1)^{r+n_1+\dots+n_r} \mathfrak{B}((n_j); (\xi_j^{-1}); (\gamma_j)).$$

zeta functions of $(-n_j) = \text{TMBNs of } (n_j)$

My work

Lemma

The following formula holds for every l, ξ .

$$\mathcal{B}_{l, \xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} = \sum_{\mathbf{e}=(e_i)_r \in \{0\} \times \{0,1\}^{r-1}} \sum_{\substack{\forall 1 \leq i \leq j \leq r \quad m_{\mathbf{e}, i, j} \in \mathbb{N}_0 \\ m_{\mathbf{e}, 1, 1} \leq l_1 \\ \vdots \\ m_{\mathbf{e}, 1, r} + \cdots + m_{\mathbf{e}, r, r} \leq l_r}} \sum_{i_1=0}^{1-e_1} \cdots \sum_{i_r=0}^{1-e_r} (-1)^{\sum_{k=1}^r (e_k + i_k)} \prod_{k=1}^r (\kappa_{k-1} + i_k)^{\sum_{j=k}^r m_{\mathbf{e}, k, j}} \prod_{k=1}^r \binom{l_k}{m_{\mathbf{e}, 1, k}, \dots, m_{\mathbf{e}, k, k}, l_k - \left(\sum_{i=1}^k m_{\mathbf{e}, i, k}\right)} \delta_{\xi, \prod_{k=1}^r \epsilon_k^{-(\kappa_{k-1} + i_k)}} \delta_{l, \sum_{k=1}^r \sum_{j=k}^r m_{\mathbf{e}, k, j}} \mathfrak{B}_{(l_1 - m_{\mathbf{e}, 1, 1} \circ_{e_2} \cdots \circ_{e_r} l_r - (\sum_{i=1}^r m_{\mathbf{e}, i, r}))} (\epsilon_1^{-1}(1 - e_1) \circ_{e_2} \cdots \circ_{e_r} \epsilon_r^{-1}(1 - e_r))$$

(a variant of CMBNs) = $\sum$ (TMBNs)

Main theorem

p-adic *L*-functions

$$= \sum (\text{a variant of CMBNs}) \times (\text{CMHVs}) \text{ [FJ]}$$

$$= \sum (\text{TMBNs}) \times (\text{CMHVs}) \text{ (F.)}$$

$$= \sum (\text{complex zeta functions}) \times (\text{CMHVs}) \text{ [FKMT]}$$

Corollary

For any $n \in \mathbb{N}$, the family

$$(p^n L_{p,1}(n; \omega^{-n}; 1; c))_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{l=0}^{\infty} \sum_{\epsilon \in \mu_c \setminus \{1\}} \binom{-n}{l} (-1)^{1+l} \zeta_1(-l; \epsilon^{-1}; 1) \mathfrak{H}(n+l, \epsilon^{-1})^{\text{Frob}^{-1}}.$$

Generalization of Bernoulli numbers

Twisted Bernoulli numbers and Bernoulli numbers

Definition

For any root of unity ξ , we define the *twisted Bernoulli numbers* by

$$\mathcal{H}(t; \xi) = \frac{1}{1 - \xi e^t} = \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!},$$

where we formally let $(-1)! = 1$.

Example

In the case $\xi = 1$, we have

$$\mathfrak{B}_{-1}(1) = -1, \quad \mathfrak{B}_n(1) = -\frac{B_{n+1}}{n+1} \quad (n \in \mathbb{N}_0).$$

In the case $\xi \neq 1$, we have

$$\mathfrak{B}_0(\xi) = \frac{1}{1 - \xi}, \quad \mathfrak{B}_1(\xi) = \frac{\xi}{(1 - \xi)^2}, \quad \mathfrak{B}_2(\xi) = \frac{\xi(\xi + 1)}{(1 - \xi)^3}, \dots$$

Twisted multiple Bernoulli numbers (TMBNs)

Definition

Let $r \in \mathbb{N}$, $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ and let $\xi_1, \dots, \xi_r \in \mathbb{C} \setminus \{1\}$ be roots of unity. Set

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) := \prod_{j=1}^r \mathcal{H} \left(\gamma_j \left(\sum_{k=j}^r t_k \right); \xi_j \right) = \prod_{j=1}^r \frac{1}{1 - \xi_j \exp(\gamma_j \sum_{k=j}^r t_k)}$$

and define *twisted multiple Bernoulli numbers* $\mathfrak{B}((n_j); (\xi_j); (\gamma_j))$ for $(n_j) \in \mathbb{Z}_{\geq 0}^r$ by

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_r=0}^{\infty} \mathfrak{B}((n_j); (\xi_j); (\gamma_j)) \frac{t_1^{n_1}}{n_1!} \cdots \frac{t_r^{n_r}}{n_r!}.$$

Generalized Euler-Zagier-Lerch type multiple zeta-function

Definition

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity. For $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$), where $\Re$ means the real part of the complex number. The *multiple zeta-function of the generalized Euler-Zagier-Lerch type* is defined by

$$\zeta_r((s_j); (\xi_j); (\gamma_j)) := \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{j=1}^r \frac{\xi_j^{m_j}}{(m_1 \gamma_1 + \cdots + m_j \gamma_j)^{s_j}},$$

which is absolutely convergent in the region

$$\mathcal{D}_r = \{(s_1, \dots, s_r) \in \mathbb{C}^r \mid \Re(s_{r-k+1} + \cdots + s_r) > k \ (1 \leq k \leq r)\}.$$

The case $\gamma_1 = \cdots = \gamma_r = 1$

Theorem

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity and $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$). Assume that

$$\xi_j \neq 1 \text{ for all } j \text{ } (1 \leq j \leq r).$$

Then, with the above notation, $\zeta_r((s_j); (\xi_j); (\gamma_j))$ can be analytically continued to $\mathbb{C}^r$ as an entire function in (s_j) . For $n_1, \dots, n_r \in \mathbb{N}_0$,

$$\zeta_r((-n_j); (\xi_j); (\gamma_j)) = (-1)^{r+n_1+\cdots+n_r} \mathfrak{B}((n_j); (\xi_j^{-1}); (\gamma_j)).$$

Remark

In the case $r = 1$, we have $\mathfrak{B}(n; \xi; 1) = \mathfrak{B}_n(\xi)$. In the general case, we use the notation $\mathfrak{B}_{(n_j)_r}((\xi_j)_r) := \mathfrak{B}((n_j)_r; (\xi_j)_r; (1)_r)$ when $\gamma_1 = \cdots = \gamma_r = 1$. Hence, we have

$$\zeta_r((-n_j); (\xi_j); (1)) = (-1)^{r+n_1+\cdots+n_r} \mathfrak{B}_{(n_j)_r}((\xi_j^{-1})_r).$$

Cyclotomic multiple harmonic sums (CMHSs)

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}. \end{aligned}$$

We define the *variant of cyclotomic multiple harmonic sums* as

$$\begin{aligned} \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \\ \sum_{\substack{(u_1, \dots, u_r) \in \mathbb{N}_0^r, \ u_1 < h \\ \forall i \geq 2, \ u_{i-1} + \kappa_{i-1} h < u_i \\ \forall i \geq 2, \ u_i < u_{i-1} + (\kappa_{i-1} + 1)h}} \left(\frac{\epsilon_2}{\epsilon_1} \right)^{u_1} \cdots \left(\frac{1}{\epsilon_r} \right)^{u_r} u_1^{l_1} \cdots u_r^{l_r} \in \mathbb{Q}(\mu_c), \end{aligned}$$

where we formally let $0^0 = 1$.

Cyclotomic multiple Bernoulli numbers (CMBNs)

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}, & \xi &\in \mu_c, \\ l &\in \{0, \dots, l_1 + \dots + l_r + r\}. \end{aligned}$$

We define the *variant of cyclotomic multiple Bernoulli numbers*

$$\mathcal{B}_{l, \xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} \in \mathbb{Q}(\mu_c)$$

by

$$\mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \sum_{\substack{0 \leq l \leq l_1 + \dots + l_r + r \\ \xi \in \mu_c}} \mathcal{B}_{l, \xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} h^l \xi^h.$$

p-adic multiple L-functions

p-adic multiple L-function

Definition

Fix a prime number p , we put $\mathcal{O}_{\mathbb{C}_p}$ to be the ring of integers of $\mathbb{C}_p$. Let $\omega : \mathcal{O}_{\mathbb{C}_p}^\times \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$ be the Teichmüller character, and let $\langle x \rangle = \frac{x}{\omega(x)}$ for $x \in \mathcal{O}_{\mathbb{C}_p}^\times$. Set

$$\int_{\mathbb{Z}_p} f(x) d\mathbf{m}_z(x) = \lim_{N \rightarrow \infty} \sum_{a=0}^{p^N-1} f(a) \mathbf{m}_z(a + p^N \mathbb{Z}_p),$$

where $\mathbf{m}_z$ is the measure defined by

$$\mathbf{m}_z(j + p^N \mathbb{Z}_p) = \frac{z^j}{1 - z^{p^N}} \quad (0 \leq j \leq p^N - 1)$$

for $z \in \mathbb{P}^1(\mathbb{C}_p)$ with $|z - 1|_p \geq 1$.

p -adic multiple L -function

Definition (continued)

Let p be a prime number and $(s_i)_r$ be the element in the set

$$\mathfrak{X}_r(d) := \{(s_1, \dots, s_r) \in \mathbb{C}_p^r \mid |s_j|_p \leq d^{-1} p^{-\frac{1}{p-1}} \ (1 \leq j \leq r)\}.$$

Let $(k_i)_r \in \mathbb{Z}^r$, $c \in \mathbb{N}_{>1}$ which is prime to p , and

$$(\mathbb{Z}_p^r)' := \{(x_1, \dots, x_r) \in \mathbb{Z}_p^r \mid p \nmid x_1, \ p \nmid (x_1 + x_2), \ \dots, \ p \nmid (x_1 + \dots + x_r)\}.$$

p -adic multiple L -function

Definition (continued)

The p -adic multiple L -function is defined by

$$L_{p,r}((s_i)_r; (\omega^{k_i})_r; (1)_r; c) :=$$

$$\int_{(\mathbb{Z}_p^r)'} \langle x_1 \rangle^{-s_1} \langle x_1 + x_2 \rangle^{-s_2} \cdots \langle x_1 + \cdots + x_r \rangle^{-s_r}$$

$$\omega^{k_1}(x_1) \omega^{k_2}(x_1 + x_2) \cdots \omega^{k_r}(x_1 + \cdots + x_r) \prod_{i=1}^r d\tilde{m}_c(x_i),$$

where $\tilde{m}_c$ is the measure defined by

$$\tilde{m}_c := \sum_{\substack{\xi^c=1 \\ \xi \neq 1}} m_\xi.$$

Expansion of special values of p -adic multiple L -function

Theorem

$$\left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c} =$$

$$\sum_{\substack{l=(l_i)_r \in \mathbb{N}_0^r \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{\substack{\epsilon \in \mu_c \\ J=(P_1, P_2, P_3) \in E_r}} \sum_{\substack{\xi \in \mu_c \\ \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2, 3)}}{\epsilon_j} \right)^{-\kappa'_j} = \xi}} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \mathcal{B}_{0, \xi}^{(l_J, \epsilon_J, \kappa_J)} \sum_{\delta \in \Delta(T_{r,J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}.$$

This infinite sum of series whose terms are $\mathbb{Q}(\mu_c)$ -linear combinations of CMHV's with depth less than or equal to r and weight tending to infinity, and whose convergence holds for the topology on $\prod_{p \in \mathcal{P}_c} \mathbb{Q}_p(\mu_c)$ of the uniform convergence with respect to $p \in \mathcal{P}_c$.

Expansion of special values of p -adic multiple L -function

Theorem

$$\begin{aligned}
 & \left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c} = \\
 & \sum_{l=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \\
 & \sum_{(e_i)_{\sharp(P_2)} \in \{0\} \times \{0,1\}^{\sharp(P_2)-1}} \sum_{i_1=0}^{1-e_1} \cdots \sum_{i_{\sharp(P_2)}=0}^{1-e_{\sharp(P_2)}} (-1)^{\sharp(P_2) + \sum_{k=1}^{\sharp(P_2)} (i_k + l_{J,k})} \\
 & \zeta_{\sharp(P_2) - \sum_{k=1}^{\sharp(P_2)} e_k} \left((-l_{J,1} \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} -l_{J,\sharp(P_2)}); \right. \\
 & \left. (\epsilon_{J,1}(1 - e_1) \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} \epsilon_{J,\sharp(P_2)}(1 - e_{\sharp(P_2)})) \right) \\
 & \delta \sum_{\delta \in \Delta(T_{r,J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}. \\
 & \epsilon \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2,3)}}{\epsilon_j} \right)^{\kappa'_j}, \prod_{k=1}^{\sharp(P_2)} \epsilon_{J,k}^{-(\kappa_{J,k-1} + i_k)}
 \end{aligned}$$

The case of depth one

TBNs and CBNs

Example (The case $r = 1$)

For any root of unity ξ_1 , TBNs is defined by

$$\mathcal{H}(t; \xi_1) = \frac{1}{1 - \xi_1 e^t} = \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi_1) \frac{t^n}{n!}.$$

Let $c \in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}$. For any $l_1 \in \mathbb{N}_0$, $\epsilon_1 \in \mu_c$, and $h \in \mathbb{N}$, CHSs is defined by

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \sum_{u_1=0}^{h-1} \epsilon_1^{-u_1} u_1^{l_1}.$$

For any $\xi \in \mu_c$ and $l \in \{0, \dots, l_1 + 1\}$, CBNs is defined by

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \sum_{\substack{l=0 \\ \xi \in \mu_c}}^{l_1+1} \mathcal{B}_{l, \xi}^{(l_1; \epsilon_1; \emptyset)} h^l \xi^h.$$

Key lemma

Lemma (The case $r = 1$)

Let $c \in \mathbb{Z}_p^\times \cap \mathbb{N}$, $c \geq 2$. Put $l_1 \in \mathbb{N}_0$ be a non-negative integer and $\epsilon_1 \in \mu_c$. Then, the following formula

$$\mathcal{B}_{l,\xi}^{(l_1;\epsilon_1;\emptyset)} = \mathfrak{B}_h(\epsilon_1^{-1})\delta_{\xi,1}\delta_{l,0} - \sum_{l'=0}^{l_1+1} \binom{l_1}{l'} \mathfrak{B}_{h-l'}(\epsilon_1^{-1})\delta_{\xi,\epsilon_1^{-1}}\delta_{l,l'}$$

holds for every ξ .

Proof.

Consider the following generating function with coefficient $\mathcal{S}_{\emptyset,h}(l_1;\epsilon_1)$

$$F(t) = \sum_{l_1=0}^{\infty} \mathcal{S}_{\emptyset,h}(l_1;\epsilon_1) \frac{t^{l_1}}{l_1!} = \sum_{l_1=0}^{\infty} \sum_{u_1=0}^{h-1} \epsilon_1^{-u_1} \frac{(u_1 t)^{l_1}}{l_1!}.$$

Key lemma

Proof (continued).

For convenience, we let $l_1 = l$, $u_1 = u$, and $\epsilon_1^{-1} = \xi$,

$$\begin{aligned} F(t) &= \sum_{l=0}^{\infty} \sum_{u=0}^{h-1} \xi^u \frac{(ut)^l}{l!} \\ &= \sum_{u=0}^{h-1} \xi^u \sum_{l=0}^{\infty} \frac{(ut)^l}{l!} \\ &= \sum_{u=0}^{h-1} \xi^u e^{ut} \\ &= \sum_{u=0}^{h-1} (\xi e^t)^u = \frac{1 - (\xi e^t)^h}{1 - \xi e^t}. \end{aligned}$$

Key lemma

Proof (continued).

By definition of TBNs, we obtain

$$\begin{aligned}
 F(t) &= \frac{1 - (\xi e^t)^h}{1 - \xi e^t} = (1 - \xi^h e^{th}) \frac{1}{1 - \xi e^t} \\
 &= \left(1 - \xi^h \sum_{l=0}^{\infty} \frac{(th)^l}{l!} \right) \left(\sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} \right) \\
 &= \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} - \left(\xi^h \sum_{l=0}^{\infty} h^l \frac{t^l}{l!} \right) \left(\sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} \right) \\
 &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l+n=m} h^l \mathfrak{B}_n(\xi) \frac{t^m}{l!n!}.
 \end{aligned}$$

Key lemma

Proof (continued).

$$\begin{aligned}
 F(t) &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l+n=m} h^l \mathfrak{B}_n(\xi) \frac{t^m}{l!n!} \\
 &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l=0}^{m+1} h^l \mathfrak{B}_{m-l}(\xi) \frac{t^m}{l!(m-l)!} \\
 &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l=0}^{m+1} h^l \mathfrak{B}_{m-l}(\xi) \binom{m}{l} \frac{t^m}{m!} \\
 &= \sum_{m=-1}^{\infty} \left(\mathfrak{B}_m(\xi) - \xi^h \sum_{l=0}^{m+1} \binom{m}{l} h^l \mathfrak{B}_{m-l}(\xi) \right) \frac{t^m}{m!}.
 \end{aligned}$$

Key lemma

Proof (continued).

By comparing the coefficients, we have

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \mathfrak{B}_h(\epsilon_1^{-1}) - \sum_{l=0}^{l_1+1} \binom{l_1}{l} \mathfrak{B}_{h-l}(\epsilon_1^{-1}) \epsilon_1^{-h} h^l.$$

By comparing the coefficients, we obtain

$$\mathcal{B}_{l, \xi}^{(l_1; \epsilon_1; \emptyset)} = \mathfrak{B}_h(\epsilon_1^{-1}) \delta_{\xi, 1} \delta_{l, 0} - \sum_{l'=0}^{l_1+1} \binom{l_1}{l'} \mathfrak{B}_{h-l'}(\epsilon_1^{-1}) \delta_{\xi, \epsilon_1^{-1}} \delta_{l, l'},$$

where δ is the Kronecker delta. □

The theorem of [FJ] and the theorem of [FKMT]

Theorem (The case $r = 1$)

$$\left(p^n L_p(n, \omega^{-n}, 1, c)\right)_{p \in \mathcal{P}_c} = \sum_{l=0}^{\infty} \binom{-n}{l} \sum_{\epsilon_1 \in \mu_c \setminus \{1\}} \sum_{\xi \in \mu_c} \mathcal{B}_{0, \xi}^{(l, \epsilon_1^{-1}, \emptyset)} \frac{\xi}{1 - \epsilon_1} \mathfrak{H}(n + l, \epsilon_1^{-1})^{Frob^{-1}}.$$

Theorem (The case $r = 1$)

$$\zeta(n; \xi; 1) = (-1)^{n+1} \mathfrak{B}_n(\xi^{-1}).$$

Main theorem

Theorem (The case $r = 1$)

$$(p^n L_p(n, \omega^{-n}, 1, c))_{p \in \mathcal{P}_c} = \sum_{l=0}^{\infty} \sum_{\epsilon \in \mu_c \setminus \{1\}} \binom{-n}{l} (-1)^{1+l} \zeta_1(-l; \epsilon^{-1}; 1) \mathfrak{H}(n+l, \epsilon^{-1})^{\text{Frob}^{-1}}.$$

Proof.

By the theorems of [FJ] and [FKMT] and the key lemma, we obtain

$$(p^n L_p(n, \omega^{-n}, 1, c))_{p \in \mathcal{P}_c} = \sum_{l=0}^{\infty} \binom{-n}{l} \sum_{\epsilon_1 \in \mu_c \setminus \{1\}} \sum_{\xi \in \mu_c} \frac{\xi}{1 - \epsilon_1} ((-1)^{l+1} \zeta(l; \epsilon_1^{-1}; 1) \delta_{\xi, 1} - (-1)^{l+1} \zeta(l; \epsilon_1^{-1}; 1) \delta_{\xi, \epsilon_1}) \mathfrak{H}(n+l, \epsilon_1^{-1})^{\text{Frob}^{-1}}.$$



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Thank you for your attention!